MILITARY SPECIFICATION

HOSE ASSEMBLY, RUBBER, LIGHTWEIGHT, COLLAPSIBLE 6 INCH:

FOR POL SERVICE

This specification is approved for use within the USA Belvoir Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a 6-inch, lightweight, collapsible rubber hose assembly for POL service; swivel, and flaking box.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

WW-C-440	- Clamps, Hose, (Low-Pressure).

MILITARY

MIL-C-104	- Crates, Wood; Lumber and Plywood Sheathed, Nailed, and Bolted.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-3774	- Crates, Wood: Open, 12,000- and 16,000-Pound Capacity.
MIL-I-6866	- Inspection, Penetrant Method of.
MIL-I-6868	- Inspection Process, Magnetic Particle.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4720

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MIL-A-8625	- Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-C-10387	- Couplings, Clamp, Pipe; with Bolts and Synthetic-Rubber Gaskets, for Grooved-End Pipe and Tube.
MIL-G-20241	- Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive.
MIL-P-23377	- Primer Coatings: Epoxy-Polyamide, Chemical and Solvent Resistant.
MIL-E-52798	- Enamel, Alkyd, Camouflage.
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWING

ME

D 13225E9182	- Flaking Box Assembly.

(Copies of specifications, standards, drawings, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B 26 - Aluminum Alloy Sand Castings.
ASTM B 221 - Aluminum Alloy, Extruded Bar, Rod, Wire, Shape and Tube.
ASTM B 247 - Aluminum and Aluminum Alloy Die and Hand Forgings.
ASTM D 380 - Rubber Hose.
ASTM D 381 - Existent Gum In Fuels by Jet Evaporation.
ASTM D 412 - Rubber Properties in Tension.
ASTM D 471 - Rubber Property-Effect of Liquids.
ASTM D 518 - Rubber Deterioration-Surface Cracking.
ASTM D 1149 - Rubber Deterioration-Surface Ozone Cracking in a Chamber (Flat Specimen).
ASTM D 3951 - Standards for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

NATIONAL SANITATION FOUNDATION STANDARD (NSF)

Plastic Piping System Components and Related Materials Standard Number 14.

(Application for copies should be addressed to the National Sanitation Foundation, NSF Building, Ann Arbor, MI 48105.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE J 1086 - Numbering Metals and Alloys.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specifications sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The hose assembly shall consist of one 500-foot section of hose. Each hose shall be fitted with grooved end adaptors on each end, and a grooved end coupling on one end. This assembly shall be packed in a flaking box. In addition, 50 percent (1 of every 2), of the hose assemblies shall be furnished with a swivel joint, so as to provide a swivel joint between every 1000 feet of hose.

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3.1.1 Drawing. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the flaking box. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3). Any changes or deviations of hose assembly from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish hose assembly conforming to this specification.

3.2.1 Sample selection plan. As part of the first article test, the contractor shall furnish a sample selection plan within the time specified (see 6.2). The plan shall assure all production hose assemblies are processed in the same manner and with production personnel as the first article hose assembly, using the same materials and methods as the first article hose assembly.

3.3 Process data. Within the time span specified (see 6.2), the contractor shall make available to the Government, on the site of manufacture, a description of materials and a documented inspection instruction which will be used by the contractor to assure that each production hose assembly is a duplicate of the first article hose assembly. Change of materials, procedures, or sources of supply from those initially selected shall be subject to the approval of the contracting officer (see 6.3 and 6.5).

3.4 Materials. Materials shall be as specified herein and on the drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.4.1 Material deterioration and control. The hose assembly shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environment to which the item may be exposed.

3.4.1.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by an insulating barrier such as a coat of epoxy primer conforming to MIL-P-23377, or corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241. Protection against any galvanic corrosion could also be obtained by exclusion of the electrolyte if feasible.

3.4.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with

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components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.4.2 Elastomer.

3.4.2.1 Tube. The basic material for the tube shall be a synthetic compound utilizing a copolymer product of butadiene and acrylonitrile, or other synthetic oil resistant compound.

3.4.2.2 Cover. The cover shall be an elastomeric or rubber compounded for fuel, oil, and ozone resistance. The color shall be sand matte, conforming to MIL-E-52798, type I.

3.4.2.3 Reinforcement material. The reinforcing material shall consist of one or more plies of high-strength synthetic fabric, evenly and firmly fabricated, free from dirt, knots, lumps, irregularities of twist, or other defects.

3.4.3 Unified numbering system (UNS). Where applicable, metal alloys are identified by UNS in accordance with SAE J 1086.

3.5 Hose assembly. The hose assembly shall consist of one 500-foot hose section with grooved-end adapters on each end and grooved end coupling on one end. (In addition, 50 percent [1 of every 2] of the hose assemblies shall be furnished with a swivel joint, so as to provide a swivel joint between every 1000 feet of hose). Each assembly shall be packed in a flaking box as shown in figure 1. The hose assembly shall have no twist when flaked in the box.

3.5.1 Hose. The hose shall consist of an inner tube, one or more plies of reinforcement material, and a protective cover. All layers shall be bonded to their adjacent layers so as to produce a unified hose wall. The color of the exterior surface of the hose cover shall approximate sand matte as defined in MIL-T-704. The hose shall be marked in white letters, 1.0 inch tall at intervals no greater than 30 feet, "POL only". The marking shall be in a straight line, so as to give the effect of a lay-line. Handling, bending, water, oil, or environmental effects shall not adversely affect the markings. Continuous branding shall be acceptable.

3.5.1.1 Dimensions and weight. The hose shall have an inside diameter of 6 inches +3/16 inch or -1/8 inch. Unless otherwise specified (see 6.2), the length of each hose section, including the two end couplings, shall be 500 feet ±5 feet. The weight of the hose shall not exceed 1.65 pounds per linear foot.

3.5.2 Hose to grooved adapter. Each end of each hose section shall be equipped with a one piece aluminum adapter with grooved end conforming to MIL-C-10387, figure 2. The aluminum shall conform to ASTM B 247, alloy UNS A96061, temper T6. Adapters shall be firmly affixed to the hose with not less than three clamps conforming to WW-C-440, type H, corrosion-resisting steel, 0.75 inch width. Use of continuous stainless steel banding shall be acceptable.

3.5.3 Grooved coupling. The grooved couplings shall conform to MIL-C-10387, class A, size 6 inch. Use of Victaulic coupling, style 77A or equal is acceptable.

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3.5.4 Swivel joint. The aluminum shall conform to ASTM B 26, alloy UNS A07120, temper F, ASTM B 221, Alloy UNS A96061, temper T6, or B 247, Alloy UNS A96061, temper T6. The swivel joint shall have grooved ends conforming to MIL-C-10387, figure 2. The joint shall be designed to permit continuous rotation at the swivel connection and shall be not more than 8 inches overall length. The joint shall be 6-inch nominal pipe size. The swivel connection shall rotate on two or more rows of ball bearings. One end shall be fitted with a grooved-end coupling.

3.5.5 Flaking box. The flaking boxes shall be in accordance with drawing D 13225E9182. The flaking boxes shall be cleaned, treated, and painted in accordance with MIL-T-704, type C, sand matte. Testing shall be as specified in 4.5.2.6. Any and all design and material changes to the flaking box design must first be approved by the contracting officer.

3.5.5.1 Proof load. The completed flaking box shall show no permanent deformation when subjected to a load not less than 6000 pounds.

3.5.5.2 Gasket. Each hose will be furnished with a gasket for 6-inch coupling, conforming to MIL-C-10387, class C, grade as specified (see 6.2), size 6-inch.

3.6 Physical properties.

3.6.1 Adhesion.

3.6.1.1 Original adhesion. The original adhesion between tube and plies, between the plies, and between the cover and the plies, shall be not less than 10 pounds.

3.6.1.2 Adhesion after immersion. A 12-inch length of hose shall be suitably stoppered and filled with reference fuel B of ASTM D 471 for 48 hours and submerged in a distilled water bath at 70° F (21° C), ±5° F (3° C). The adhesion between the tube and plies, between the plies, and between the cover and the plies shall be not less than 6 pounds.

3.6.1.3 Low temperature flexibility. When tested as specified in 4.5.2.4.6, there shall be no evidence of cracking, lack of flexibility, or any other defects.

3.6.2 Tensile strength. The tensile strength of the tube shall be not less than 1,250 pounds per square inch (psi) and the tensile strength of the cover shall be not less than 1,500 psi. The tensile strength of the tube and cover (after immersion), shall be not less than 40 percent or less than 600 psi (whichever is greater) of the tensile strength obtained before immersion.

3.6.3 Ultimate elongation. The ultimate elongation of the tube and cover shall be not less than 300 percent, and the elongation after immersion shall be not less than 200 percent.

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3.6.4 Ozone resistance. The hose cover shall withstand 50 parts of ozone per 100 million parts of air (pphm), maintained at +104° F (+40° C) for 168 hours without evidence of cracking when observed under 7-power magnification.

3.6.5 Existent gum content. When tested as specified in 4.5.2.3.1, the remainder shall be not more than 20 milligrams per 100 milliliters of test fluid.

3.6.6 Hydrostatic pressure. The hose assembly shall withstand an internal hydrostatic pressure of not less than 225 psi without rupture or leak (see 6.4), either in the hose body or around the couplings and without any deformation of the couplings.

3.6.7 Elongation and twist. The hose section (500-foot), when subjected to an internal hydrostatic pressure of 150 psi shall not show elongation greater than one percent in length; or twist, either right or left, by more than 1/2 turn per hose section when tested as specified in 4.5.2.2.2.

3.6.8 Burst pressure. The hose shall withstand a burst pressure of not less than 500 psi without rupture or failure (see 6.4.2).

3.7 Identification marking. The hose assembly shall be identified in accordance with MIL-STD-130. Each hose assembly shall have at approximately 50-foot intervals the following information:

Date of manufacture (quarter and year).
Contract or order number.
Manufacturer's name and trademark.

3.8 Metallic component finish. Unless otherwise specified herein, aluminum parts shall have a dyed, anodized coating conforming to MIL-A-8625, type II, class 2, color to approximate sand matte as defined in MIL-T-704. The anodic coating shall be not less than 0.0007 inch thick.

3.9 Workmanship. The hose assembly shall have no defects affecting appearance or serviceability, or safety. The hose shall be of uniform quality and shall be free from cracks, pitting, or other defects.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The

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absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article hose assembly and flaking box shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article hose assembly and flaking box shall be subjected to the tests marked "X" in column 1 of table I. Samples for adhesion tests shall be cut from a length of hose. Samples of the synthetic rubber compound shall be representative of materials used to fabricate the hose assembly. Failure of any test shall be cause for rejection of the hose assembly or for performing the flaking box disassembly inspection specified in 4.1.2 if the defects are found in the flaking box.

4.4 Quality conformance inspection.

4.4.1 Sampling.

4.4.1.1 Examination. Sampling of hose assemblies for examination shall be in accordance with MIL-STD-105, normal, level II, AQL 2.5 percent defective.

4.4.1.2 Tests.

4.4.1.2.1 Elastomeric or rubber compound. Four test slabs, each 6 inches by 6 inches by .075 inch, ±.010 inch thick, of the tube and cover compounds and of equivalent cure used to fabricate the hose assemblies, shall be supplied with each lot.

4.4.1.2.2 Hose. Sampling of hose for tests shall be in accordance with MIL-STD-105, inspection level S-2. AQL shall be 2.5 percent defective. Test sections shall be cut from the hose and shall be sufficient length for the performance of the required tests.

4.4.1.2.3 Flaking boxes. Sampling of flaking boxes for tests shall be in accordance with MIL-STD-105, inspection level S-2. AQL shall be 4.0 percent defective.

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4.4.2 Examination. Sample hose assemblies selected in accordance with 4.4.1.1 shall be examined as specified in 4.5.1.

4.4.3 Tests.

4.4.3.1 Individual. Each hose assembly shall be subjected to the test marked "X" in column 2 of table I. Failure of the tests shall be cause for rejection.

4.4.3.2 Samples. Samples selected in accordance with 4.4.1.2.1, 4.4.1.2.2, and 4.4.1.2.3 shall be subjected to the tests marked "X" in column 3 of table I.

4.5 Inspection procedure.

4.5.1 Examination. The hose assembly and flaking box shall be examined as specified herein for the following defects:

101. Material not as specified.
102. Materials are not corrosion resistant or treated to be made corrosion resistant for the applicable storage and operating environment.
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
104. Contractor does not have documentation available for identification of material, material finishes, or treatments.
105. Hose twisted when flaked in boxes.
106. Hose not as specified.
107. Dimensions and weight not as specified.
108. Hose to grooved adapter not as specified.
109. Clamps not as specified.
110. Grooved couplings not as specified.
111. Swivel joint not as specified.
112. Hose adapter and flaking boxes not as specified.
113. Identification marking missing, incomplete, or illegible.
114. Finish not as specified.
115. Workmanship not as specified.
116. Metal alloys not identified by UNS.
117. Gasket missing or not as specified.
118. Elastomer not as specified.
119. Process data missing or not as specified.
120. Sample selection plan missing or not as specified.
121. Physical properties not as specified.
122. Color not as specified.

4.5.2 Tests.

4.5.2.1 Test schedule.

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TABLE I. Test schedule.

First article	Quality conformance: Individual	Quality conformance: Sample	Test	Test paragraph	Requirement paragraph
1	2	3	4	5	6
			Hose assembly.		
X	X	-	Hydrostatic	4.5.2.2.1	3.6.6
X	X	X1/	Elongation and twist.	4.5.2.2.2	3.6.7
			Elastomeric or rubber compounds.		
X	-	X	Initial tensile strength.	4.5.2.3	3.6.2
X	-	X	Tensile strength after immersion.	4.5.2.3	3.6.2
X	-	X	Ultimate elongation, initial.	4.5.2.3	3.6.3
X	-	X	Ultimate elongation after immersion.	4.5.2.3	3.6.3
X	-	X	Ozone resistance.	4.5.2.3	3.6.4
			Hose.		
X	-	X	Burst pressure.	4.5.2.4.1	3.6.8
X	-	X	Weight.	4.5.2.4.2	3.5.1.1
X	-	X	Original adhesion.	4.5.2.4.3	3.6.1.1
X	-	X	Adhesion after immersion in fuel.	4.5.2.4.4	3.6.1.2
X	-	X	Adhesion after immersion in water.	4.5.2.4.5	3.6.1.2
X	-	X1/	Low temperature flexibility.	4.5.2.4.6	3.6.1.3
X	-	X	Existent gum.	4.5.2.3.1	3.6.5
			Flaking boxes.		
X	-	X	Proof load.	4.5.2.5	3.5.5.1

1/ One lot testing required for each ten lots submitted.

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4.5.2.2 Hose assembly.

4.5.2.2.1 Hydrostatic pressure. Each hose section shall be subjected to the hydrostatic tests, using water as a test fluid, of ASTM D 380 for proof pressure. Nonconformance to 3.6.6 shall constitute failure of this test. The hydrostatic test shall be performed with water. After testing, the hose shall be dried and the ends shall be sealed.

4.5.2.2.2 Elongation and twist. Elongation and twist measurement shall be on a hose length of 500 feet ±5 feet, laid out straight. One end of the hose shall be fixed and not allowed to turn. Twist shall be measured at the free end and recorded as clockwise or counterclockwise. Pressurize the hose section, using water as a test fluid, to 10 psi and hold for not more than two minutes and mark the length and twist position. After two minutes increase the pressure to 150 psi and mark the length and twist position. Nonconformance to 3.6.7 shall constitute failure of this test.

4.5.2.3 Elastomeric or rubber compounds. Elastomeric or rubber compound sample shall be tested as follows:

Property	Test method
Initial tensile strength.	ASTM D 412
Tensile strength after immersion.	ASTM D 471 2/
Ozone resistance.	ASTM D 1149 1/
Initial ultimate elongation.	ASTM D 412
Ultimate elongation after immersion.	ASTM D 471 2/

1/ Specimens mounted in accordance with procedure B, ASTM D 518, ozone concentration shall be 50 ±10 parts per 100,000,000 (pphm). Time of exposure shall be 168 hours; temperature of exposure shall be 104° F (40° C).

2/ The samples shall be immersed in reference fuel B for 14 days at 158° F (70° C), ±3.6° F (2° C).

Nonconformance to 3.6.2, 3.6.3, or 3.6.4 shall constitute failure of these tests.

4.5.2.3.1 Existent gum. A test sample of not less than 14 inches long shall be cut from a test section of hose selected under 4.4.1.2.2. The bottom end of the sample shall be stoppered with a clean noncorrosive plug 2 inches long secured in the end of the sample with a suitable clamp. The sample shall then be filled to within 2 inches from the top with ASTM D 471 reference fuel D, and the top end plugged in a manner similar to the bottom. This sample shall then be stored in a vertical position for 7 days at an ambient temperature of 100° F (37.8° C) ±2° F (1° C). At the end of each 24 hours the fuel shall be agitated for 5 minutes by moving the hose from the vertical to horizontal position at a rate of two cycles per minute. At completion of the 7 days storage period, fuel shall again be agitated in this same manner for 5 minutes and immediately removed from the hose. The removed fuel shall be tested for existent gum in

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accordance with the air-jet solvent wash method of ASTM, D 381. A sample of test fuel before entering the hose shall be tested at the same time and by the same method. The existent gum of this sample shall be subtracted from the existent gum obtained from the fuel removed from the test hose. Nonconformance to 3.6.5 shall constitute failure of this test.

4.5.2.4 Hose.

4.5.2.4.1 Burst pressure. The test section of hose selected under 4.4.1.2.2 shall be subjected to the straight bursting test of ASTM D 380. Nonconformance to 3.6.8 shall constitute failure of this test.

4.5.2.4.2 Weight. A section 12 inches long shall be cut from the test section of hose selected under 4.4.1.2.2 and weighed. Nonconformance to 3.5.1.1 shall constitute failure of this test.

4.5.2.4.3 Original adhesion. From the test section of hose selected under 4.4.1.2.2 prepare ring specimens as described in ASTM D 380. The adhesion of the ring specimen shall be determined in accordance with ASTM D 380, machine method. Nonconformance to 3.6.1.1 shall constitute failure of this test.

4.5.2.4.4 Adhesion after immersion in fuel. Specimens prepared as described in 4.5.2.2.2 shall be immersed as described in 3.6.1.2. Within 5 minutes after removal from immersion, the adhesion shall be determined as described in 4.5.2.2.2. Nonconformance to 3.6.1.2 shall constitute failure of this test.

4.5.2.4.5 Adhesion after immersion in water. Specimens prepared as described in 4.5.2.2.2 shall be immersed as described in 3.6.1.2. Within 15 minutes after removal from immersion, the adhesion shall be determined as described in 4.5.2.2.2. Nonconformance to 3.6.1.2 shall constitute failure of this test.

4.5.2.4.6 Low temperature flexibility. A sample 8-foot length of hose shall be folded 180 degrees and tied in the folded position. The hose shall then be conditioned for 72 hours at -25° F (-31.7° C). The hose shall be removed from the test chamber and unfolded 180 degrees using a force of not more than 10 pounds within 5 seconds. The unfolding shall be accomplished within 3 minutes after removal from the test chamber. The hose sample shall then be hydrostatic pressure tested at 225 psig in accordance with ASTM D 380. Hose fabricated with splices shall have two hose test samples, one hose sample folded without a splice, and the other sample folded at the splice. Nonconformance of any test sample to 3.6.1.3 shall constitute failure of this test.

4.5.2.5 Proof load. Distribute a minimum 6,000 pound load in one or more flaking boxes as necessary. The boxes shall be filled and stacked one on top of the other until the total specified load of 6,000 pounds is achieved. Attach a lifting sling to the bottom flaking box and lift boxes to a height of not less than 1 foot. After the lifted box comes to a complete rest, hold load for not less than 5 minutes. Lower flaking boxes to the ground and inspect all welding and lifting eyes on the bottom flaking box by one of the following methods after the test:

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a. Magnetic particle in accordance with MIL-I-6868. Defects will not be permitted.
b. Dye penetrant inspection in accordance with MIL-I-6866, type I, method B. Defects will not be permitted.

Nonconformance to one of the above using either welding inspections or 3.5.5.1 shall constitute failure of the test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105, normal level II, AQL 2.5 percent defective.

4.6.1.3 Examination. Samples selected for examination in accordance with 4.6.1.2 shall be examined for applicable defects listed in table II.

TABLE II. Examination.

No.	Defect	A	B	Comm.
121.	More than four flaking boxes packed per crate.	5.1.1	5.1.2	
122.	Exterior containers not as specified.	5.1.1	5.1.2	5.1.3
123.	Contents not shrouded as specified.		5.1.2	
124.	Blocking, bracing, anchoring, closure, and strapping not as specified.	5.1.1	5.1.2	
125.	Hold down devices not as specified (if required).	5.1.1	5.1.2	
126.	Marking not as specified.	5.2.1	5.2.1	5.2.2

5. PACKAGING

5.1 Packing. Packing shall be level A, level B, or commercial, as specified (see 6.2).

5.1.1 Level A. The flaking boxes, with the hose assemblies positioned inside in the manner specified in 3.5 (see figure 1), shall be packed in quantities not to exceed four boxes, in a crate conforming to MIL-C-104, type I, class 1 or 2, style A. Blocking, bracing and anchoring of the boxes within the crate shall be in accordance with the appendix to the specification. Additionally, held down devices of adequate design and quantity shall be utilized as required, to reduce to a minimum, the vertical movement of the hose assembly with the flaking box.

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Non-abrasive cushioning material shall be used between the hold down devices and the hose assemblies, as required, to prevent damage to the hose. Hold down devices shall be secured in position.

5.1.2 Level B. The flaking boxes, with the hose assemblies positioned inside in the manner specified in 3.5 (see figure 1), shall be packed, in quantities not to exceed four boxes, in a crate conforming to MIL-C-3774, type I, style A. The contents shall be protected with a waterproofed shroud and its application shall be in accordance with MIL-STD-1186. The crate closure and strapping shall be in accordance with the appendix to the crate specification. Additionally, the application of the hold down devices specified in 5.1.1 shall be utilized, as required, to immobilize and protect the hose assemblies.

5.1.3 Commercial. The flaking boxes, with the hose assemblies positioned inside in the manner specified in 3.5 (see figure 1), shall be packed in accordance with ASTM D 3951.

5.2 Marking.

5.2.1 Levels A and B. In addition to any special or other identification marking required by the contract (see 6.2), each crate shall be marked in accordance with MIL-STD-129.

5.2.2 Commercial. Commercial marking shall be in accordance with the requirements of ASTM D 3951. Each shipping container shall be marked with the cube and gross weight.

6. NOTES

6.1 Intended use. The hose assembly is intended for use with light-weight, collapsible-hose pipelines in fuel systems.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When a first article is required for inspection and approval and the number of units required (see 3.2).
c. Time frame required for submission of sample selection plan and process data (see 3.2.1 and 3.3).
d. Length of hose if other than 500 feet (see 3.5.1.1).
e. Grade of gasket required (see 3.5.5.2).
f. Degree of packing required (see 5.1).
g. Any special marking instructions (see 5.2.1).

6.3 First article. When a first article inspection is required, the items will be tested and should be a first produced model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the first article test results and disposition of the document's first article.

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6.4 Definitions.

6.4.1 Leak. A leak is defined as any passage of fluid from the inner portion of the hose assembly to the outer portion of the hose assembly as determined by sight or touch. A leak shall include but is not limited to fluid passing through a hose assembly fitting, fluid passing through the hose body, or fluid passing through the junction of the hose and an attached fitting.

6.4.2 Rupture. A rupture in the hose assembly is defined as a split or crack in the hose, or any split or crack at the junction of hose and a fitting.

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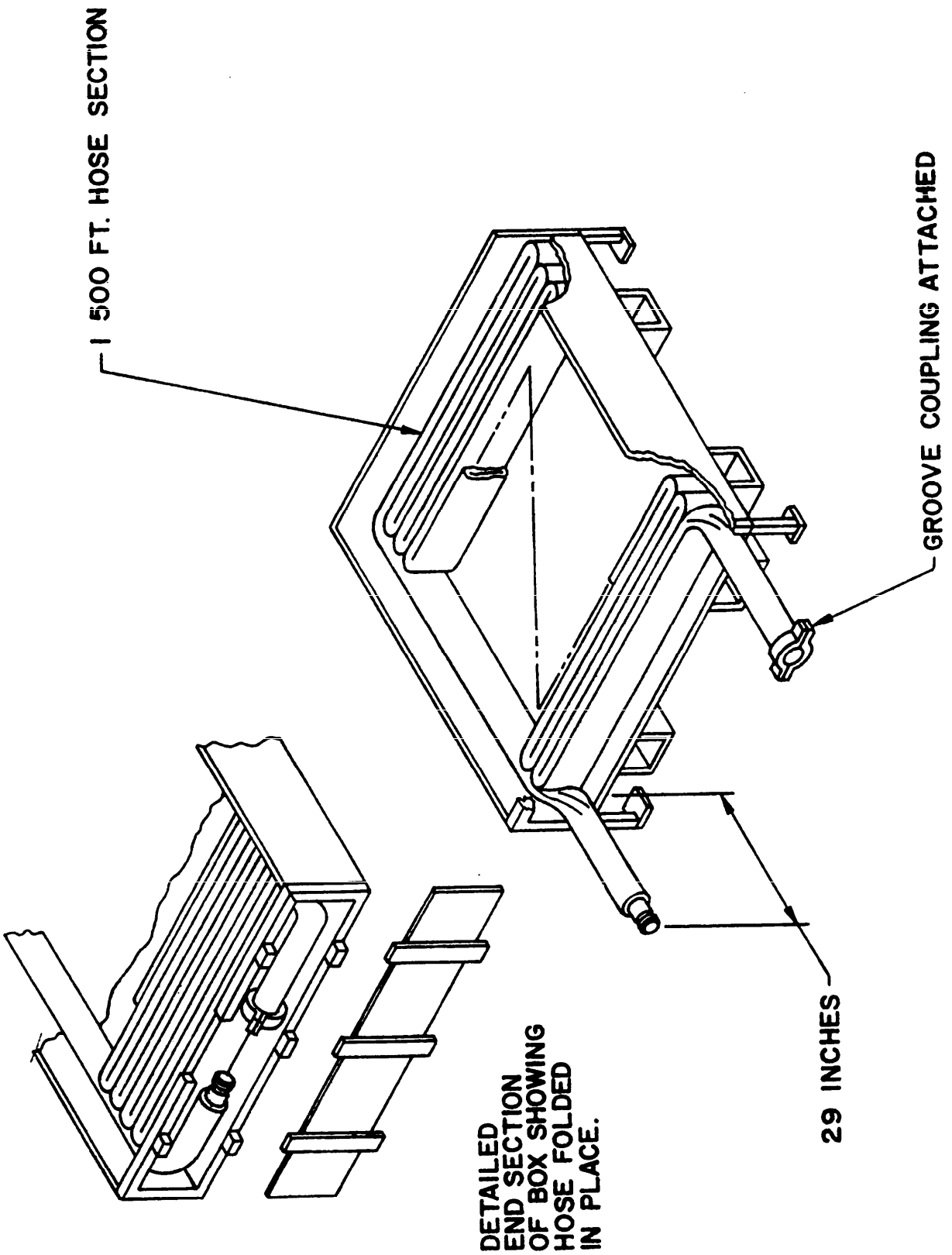


FIGURE 1. Method of packing hose in flaking boxes.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-H-53062(ME)	Hose Assembly, Rubber, Lightweight, Collapsible 6 Inch: for POL Service

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: ____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include A Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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